

Question number	Answer	Additional guidance	Mark
5(a)	An explanation that includes the following points: (safe / not too high a dose) to avoid {uncontrollable bleeding / named examples of uncontrollable bleeding / internal bleeding} (1) (effective / not too low a dose) to reduce (the risk of) blood clotting (1)	IGNORE side effects / harm / toxicity ACCEPT passing blood in your urine passing blood in faeces or having black faeces severe bruising prolonged nosebleeds vomiting or coughing up blood heavy or increased bleeding during periods, or any other bleeding from the vagina jaundice ACCEPT thin the blood	(2)

Question number	Answer	Additional guidance	Mark
5(b)(i)	A description that includes the following points: - to carry a <u>copy</u> of the gene {out of the nucleus / into the cytoplasm / to the ribosomes} (1) (used for) {(correct) ordering of the amino acids / producing amino acid sequence / formation of (poly)peptide chain} (1)	ACCEPT {(genetic) code / DNA base sequence / DNA} <u>if reference to the enzyme has been made</u> NB copy is the noun not the verb	(2)

Question number	Answer
*5(b)(ii)	(6) EXP Advantages of pharmacogenomics: • genotype {affects / determines} the dose of warfarin needed • so one dose will not be appropriate for all patients • as the two genes that have several alleles • (one / two) gene M allele that slows down breakdown means less warfarin - accept converse for other alleles • one gene V allele results in less enzyme V / mRNA so less warfarin should be prescribed - accept converse for other alleles • other genetic predispositions maybe identified e.g. high cholesterol, high blood pressure Effect of other factors: • factors other than genotype affects blood clotting so these would need to be taken into account as well • e.g. people with high cholesterol more likely to be prescribed warfarin • mass of warfarin would need to be adjusted according to body size / given at a particular dose in mgs per kg • liver function may affect the half life of the drug Issues surrounding genetic screening: • there are religious / moral / social / ethical issues relating to genetic screening • people / family may not want to find out if there are any other underlying genetic issues • possibility of false negative / positive results Blood clotting: • thromboplastin converts prothrombin into thrombin which converts fibrinogen into fibrin to trap platelets

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Simple points made 1 mark = 1 point made 2 marks = some attempt to comment on pharmacogenetics or factors that affect it OR description of blood clotting process
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Simple discussion of pharmacogenetics 3 marks = implications of genotype for either gene M or gene V on warfarin dose 4 marks = implications of genotype for both gene M and gene V on warfarin dose
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Understanding of other issues involved 5 marks = criteria for 4 marks plus links made with either blood clotting process or other factors or issues of screening 6 marks = criteria for 4 marks plus links made with two from blood clotting process / other factors / issues surrounding screening